Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

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Contents

Chapter 0

Preliminaries

0.2 Quantifiers and Negation

Exercise 0.2.1 Negate the following statements:

a. Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.

b. For all $x \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

c. For all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

SOLUTION

a. There exists a prime number in the form of $4n+1$, for $n \in \mathbb{N}_0$ that cannot be expressed as the sum of two squares.

b. There exists $x \in \mathbb{R}$ and $\epsilon > 0$ such that for all $\delta > 0$, there exists $y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

c. There exists $\epsilon > 0$ such that for all $\delta > 0$, there exists $x, y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

Exercise 0.2.2 Explain why one of these statements is true and the other is false:

$$(\forall \text{ man } M)(\exists \text{ woman } W) \mid W \text{ is the mother of M}$$
$$(\exists \text{ woman } W)(\forall \text{ man } M) \mid W \text{ is the mother of M}$$

SOLUTION

The difference between the two statements lies in the quantifier. Obviously the first statement is true while the second is false.

The first statement states that for every man, we can find a woman that is the man's father. This is inherently true as every man must have a mother.

On the other hand, the second statement states that we can find a women that is the mother of every man, which is obviously not possible.

0.3 Set Theory

Exercise 0.3.1 Let E be a set, with subsets $A \subset E$ and $B \subset E$, and let $*$ be the operation $A * B = (E-A) \cap (E-B)$. Express the sets

$$\text{a. } A \cup B \qquad \text{b. } A \cap B \qquad \text{c. } E - A$$

using A, B and $*$.

SOLUTION

Before solving this problem, consider the special case of the operator $*$:

$$X * X = (E-X) \cap (E-X) = X^c \cap X^c = X^c$$

a. We can first simpify the operator with $A * B = A^c \cap B^c = (A \cup B)^c$. Using the special case, we get

$$(A * B) * (A * B) = (A \cup B)^c * (A \cup B)^c = A \cup B$$

b. Working backwards, we get

$$A \cap B = (A^c \cup B^c)^c = ((A * A) \cup (B * B))^c = (A * A) * (B * B)$$

c. This case is the equivilant to the special case above, hence $E - A = A^c = A * A$.

0.4 Functions

Exercise 0.4.1 Are the following true functions? That is, are they both everywhere defined and well defined?

a. "The aunt of", from people to people.

b. $f(x) = \frac{1}{x}$, from real numbers to real numbers.

c. "The capital of", from countries to cities (careful - at least one country, Bolivia, has to capitals.)

SOLUTION

a. Considering the cases where and individual's parents not having any sisters, the individual does not have an aunt, making it not everywhere defined. Considering the cases where an individual has multiple aunts, the function is not well defined.

b. The function is not everywhere defined on real numbers as when $x = 0$, $f(x)$ is undefined. The function however is wel defined as if $f(x) = f(y)$, then $x = y$.

c. The function is everywhere defined as every country has a capital. However, it is not well defined as some countries, such as Bolivia, has two capitals.

Exercise 0.4.2 a. Make up a nonmathematical function that is onto but not one to one.

b. Make up a mathematical function that is onto but not one to one.

SOLUTION

a. The suit of a poker card is onto but not one to one; every card has a suit but more that one card has the same suit.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x^2$

Exercise 0.4.3 a. Make up a nonmathematical function that is bijective (onto and one to one).

b. Make up a mathematical function that is bijective.

SOLUTION

a. The thumbprint of a person is bijective as everyone has a thumbprint that is unique.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x$.

Exercise 0.4.4 a. Make up a nonmathematical function that is one to one but not onto.

b. Make up a mathematical function that is one to one but not onto.

SOLUTION

a. The function mapping credit card to credit card numbers is one to one but not onto; as each card number is unique but not all card numbers have been used.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = e^x$.

Exercise 0.4.5 Given the functions $f : A \to B$, $g : B \to C$, $h : A \to C$, and $k : C \to A$, which of the following compositions are well defined? For those that are, give the domain and codomain of each.

a.	$f \circ g$	b.	$g \circ f$	c.	$h \circ f$	d.	$g \circ h$	e.	$k \circ h$	f.	$h \circ g$
g.	$h \circ k$	h.	$k \circ g$	i.	$k \circ f$	j.	$f \circ h$	k.	$f \circ k$	l.	$g \circ k$

SOLUTION

a.	Not well defined.	b.	$g \circ f : A \to C$	c.	Not well defined.	d.	Not well defined.
e.	$k \circ h : A \to A$	f.	Not well defined.	g.	$h \circ k : C \to C$	h.	$k \circ g : B \to A$
i.	Not well defined.	j.	Not well defined.	k.	$f \circ k : C \to B$	l.	Not well defined.

Exercise 0.4.6 Prove Proposition 0.4.11. **Inverse image of intersection, union**

a. The inverse image of an intersection equals the intersection of the inverse images:

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

b. The inverse image of a union equals the union of the inverse images:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

SOLUTION

a. We will show this by inclusion both ways:

   i. Let $y \in A \cap B$ with $x = f^{-1}(y) \in f^{-1}(A \cap B)$. Then $y \in A$ and $y \in B$, implying $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cap f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cap f^{-1}(B)$, then $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, so $y \in A$ and $y \in B$. This shows that $y \in A \cap B$, hence, $x \in f^{-1}(A \cap B)$.

b. Similarly, by inclusion both ways:

   i. Let $y \in A \cup B$ with $x = f^{-1}(y) \in f^{-1}(A \cup B)$. Then $y \in A$ or $y \in B$, implying $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cup f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cup f^{-1}(B)$, then $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, so $y \in A$ or $y \in B$. This shows that $y \in A \cup B$, hence, $x \in f^{-1}(A \cup B)$.

Exercise 0.4.7 Evaluate $(f \circ g \circ h)(x)$ at $x = a$ for the following.

a. $f(x) = x^2 - 1, \quad g(x) = 3x, \quad h(x) = -x + 2, \quad$ for $a = 3$.

b. $f(x) = x^2, \quad g(x) = x - 3, \quad h(x) = x - 3, \quad$ for $a = 1$

SOLUTION

a.

$$\begin{aligned}(f \circ g \circ h)(3) &= f(g(h(3))) \\ &= f(g(-3+2)) = f(g(-1)) \\ &= f(3(-1)) = f(-3) \\ &= (-3)^2 - 1 = 8\end{aligned}$$

b.

$$\begin{aligned}(f \circ g \circ h)(1) &= f(g(h(1))) \\ &= f(g(1-3)) = f(g(-2)) \\ &= f(-2-3) = f(-5) \\ &= (-5)^2 = 25\end{aligned}$$

Exercise 0.4.8 a. Prove Proposition 0.4.16. **(Composition of onto functions).** Let the functions $f : B \to C$ and $g : A \to B$ be onto. Then the composition $(f \circ g)$ is onto.

b. Prove Proposition 0.4.17. **(Composition of one to one functions).** Let $f : B \to C$ and $g : A \to B$ be one to one. Then the composition $(f \circ g)$ is one to one.

SOLUTION

a. Let $c \in C$. Since f is onto, there exists some $b \in B$ such that $f(b) = c$. Since g is onto, there exists some $a \in A$ such that $g(a) = b$. Hence $(f \circ g)(a) = f(g(a)) = f(b) = c$. Since every $c \in C$ has a preimage under $f \circ g$, the composition $f \circ g$ is onto.

b. Let $a_1, a_2 \in A$ and suppose $(f \circ g)(a_1) = (f \circ g)(a_2)$. Then $f(g(a_1)) = f(g(a_2))$. Since f is one to one, $g(a_1) = g(a_2)$. Since g is one to one, $a_1 = a_2$. Therefore $f \circ g$ is one to one.

Exercise 0.4.9 What is the natural domain of $\sqrt{\dfrac{x}{y}}$?

SOLUTION

For $\sqrt{\frac{x}{y}}$ to be defined, $\frac{x}{y} \geq 0$. Hence the natural domain is $\{(x, y) \in \mathbb{R}^2 : y \neq 0, \frac{x}{y} \geq 0\}$.

Exercise 0.4.10 What is the natural domain of

a. $\ln \circ \ln$? b. $\ln \circ \ln \circ \ln$? c. $\ln$ composed with itself n times?

SOLUTION

a. The domain of $\ln x$ is $x > 0$. Hence the domain of $\ln \circ \ln$ is when $\ln x > 0$ which is $\{x \in \mathbb{R} : x > 1\}$.

b. Similarly, the natural domain of $\ln \circ \ln \circ \ln$ is $\{x \in \mathbb{R} : x > e\}$.

c. The natural domain of $\ln$ composed with itself n times is $\{x \in \mathbb{R} : x > e \uparrow\uparrow (n - 2)\}$.

Exercise 0.4.11 What subset of $\mathbb{R}$ is the natural domain of the function $(1 + x)^{1/x}$?

SOLUTION

For $(1 + x)^{1/x}$ to be defined, we must have the following restrictions, $(1 + x) > 0$; to avoid fractional powers of negative numbers, and $x \neq 0$; as we cannot divide by zero. Hence, the natural domain is $(1, \infty) \setminus \{0\}$.

Exercise 0.4.12 The function $f(x) = x^2$ from real numbers to real nonnegative numbers is onto but not one to one.

a. Can you make it one to one by changing its domain? By changing its codomain?

b. Can you make it not onto by changing its domain? Its codomain?

SOLUTION

a. Yes.

   i. Change the domain to $[0, \infty)$. Then $f(x) = x^2$ is one to one and remains onto $\mathbb{R}_{\geq 0}$.

   ii. Changing the codomain to $\{0\}$, makes $f(x) = x^2$ trivially one to one

b. Yes.

   i. Change the domain to $(0, \infty)$. Then 0 is no longer attained, so f is not onto $\mathbb{R}_{\geq 0}$.

   ii. Change the codomain to $\mathbb{R}$. Then negative real numbers are not attained, so f is not onto.

0.5 Real Numbers

Exercise 0.5.1 Show that if p is a polynomial of odd degree with real coefficients, then thereis a real number c such that $p(c) = 0$.

SOLUTION
We will begin by finding two values to contruct the requirements for Intermediate Value Theorem. Firstly, let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$. Without loss of generality, suppose $a_n > 0$, then $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = \infty$; since n is odd. Hence, we can find $N > 0$ large enough such that $p(-N) < 0$ and $p(N) > 0$. By Intermediate Value Theorem, we can find $c \in (-N, N)$ such that $p(c) = 0$.

Exercise 0.5.2 a. Show that the function

$$f(x) = \begin{cases} \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ is not continuous.}$$

b. Show that f satisfies the conclusion of the intermediate value theorem: If $f(x_1) = a_1$ and $f(x_2) = a_2$, then for any number a between a_1 and a_2, there exists a number x between x_1 and x_2 such that $f(x) = a$.

SOLUTION

a. Intuitively, the discontinuity must occur at $x = 0$ since $1/x$ has a discontinuity at the same point. To prove this, we must find a subsequences to show that the limit does not agree at zero. Consider

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n} \to 0 \implies f(x_n) = \sin\left(\frac{1}{x_n}\right) = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1 \neq f(0)$$

Since $\lim_{x\to 0} f(x) \neq f(0)$, f is not continuous.

b. Since f is continuous on $\mathbb{R} \setminus \{0\}$, if both $x_1, x_2 < 0$ or $x_1, x_2 > 0$, then Intermediate Value Theorem applies as it will be continuous on the interval $[x_1, x_2]$ or $[x_2, x_1]$. Hence we just have to prove this for the cases where $x_1 x_2 < 0$.

Without loss of generality, suppose $x_1 < 0 < x_2$. For $\epsilon > 0$, we can find $N > 0$ big enough such that

$$\frac{(4N+3)\pi}{2} > \frac{(4N+1)\pi}{2} > 1/\epsilon$$

Let $\epsilon < \min\{|x_1|, |x_2|\}$. Then we can find an interval $\left[\frac{2}{(4N+3)\pi}, \frac{2}{(4N+1)\pi}\right] \subseteq [x_1, x_2]$. Since

$$f\left(\frac{2}{(4N+1)\pi}\right) = -1, \quad f\left(\frac{2}{(4N+3)\pi}\right) = 1$$

Intermediate Value Theorem applies, thus we can find $f(x) = a$ for $a \in [a_1, a_2] \subseteq [-1, 1]$; since the range of $\sin(1/x)$ is $[-1, 1]$.

Exercise 0.5.3 Suppose $a \leq b$. Show that if $f : [a, b] \to [a, b]$ is continuous, there exists $c \in [a, b]$ with $f(c) = c$.

SOLUTION
We consider the function $g(x) = f(x) - x$. Then

$$f(x) \in [a, b] \implies a \leq f(x) \leq b$$

so

$$f(a) \geq a \implies g(a) = f(a) - a \geq 0$$

and

$$f(b) \leq b \implies g(b) = f(b) - b \leq 0$$

Since $g(a)g(b) \leq 0$ and g is continuous on $[a, b]$, Intermediate Value Theorem states we can find a $c \in [a, b]$ such that $g(c) = 0$, which means $f(c) = c$.

Exercise 0.5.4 Let
$$a_n = \frac{(-1)^{n+1}}{n}, \quad \text{for } n = 1, 2, \ldots$$

a. Show that the series $\sum a_n$ is convergent.

b. Show that $\sum_{n=1}^{\infty} a_n = \ln 2$.

c. Explain how to rearrange the terms of the series so it converges to 5.

d. Explain how to rearrange the terms of the series so that it diverges.

SOLUTION

a. We will begin by grouping the terms so that we can apply the Monotone Convergence Theorem **(Theorem 0.5.7)**.

$$s_{2n+1} = \sum_{i=1}^{2n+1} a_i = 1 + \sum_{i=1}^{n} a_{2i+1} - \sum_{i=1}^{n} a_{2i} = 1 + \sum_{i=1}^{n} (a_{2i+1} - a_{2i}) = 1 + \sum_{i=1}^{n} \left(\frac{1}{2i+1} - \frac{1}{2i} \right)$$

Since $\frac{1}{2i+1} < \frac{1}{2i}$, then $\frac{1}{2i+1} - \frac{1}{2i} < 0$. So s_{2n+1} is strictly decreasing, and $s_{2n+1} < 1$ for all $n \in \mathbb{N}$. Similarly,

$$s_{2n} = \sum_{i=1}^{2n} a_i = \sum_{i=1}^{n} a_{2i-1} - \sum_{i=1}^{n} a_{2i} = \sum_{i=1}^{n} (a_{2i-1} - a_{2i}) = \sum_{i=1}^{n} \left(\frac{1}{2i-1} - \frac{1}{2i} \right)$$

Since $\frac{1}{2i-1} > \frac{1}{2i}$, then $\frac{1}{2i-1} - \frac{1}{2i} > 0$. So s_{2n} is strictly increasing, and $s_{2n} > 0$ for all $n \in \mathbb{N}$.
Using $0 < s_{2n} < s_{2n+1} < 1$ and by the Monotone Convergence Theorem, (s_{2n+1}) is monotone decreasing and bounded below, thus converging. Similarly, (s_{2n}) is monotone increasing and bounded above, thus converging.

We have shown that the subsequences of odd terms and even terms converge, now we have to show that they converge to the same value. Suppose $(s_{2n}) \to M$ and $(s_{2n+1}) \to L$, then taking limits on both sides of

$$\begin{aligned}
|s_{2n+1} - s_{2n}| &= \left| \frac{1}{2n+1} \right| \\
\implies |M - L| &= |\lim a_{2n+1} - s_{2n}| \\
&= \lim \left| \frac{1}{2n+1} \right| = 0
\end{aligned}$$

showing that both subsequence converge to a single value.

b. Consider the Maclaurin Expansion of $\ln(1 + x)$

$$\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad \text{for } |x| \leq 1$$

Substituting $x = 1$, we get

$$\ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \sum_{n=1}^{\infty} a_n$$

c. We can construct a series that converges to 5 following this algorithm. If the sum is under 5, the next term in the series will be the next unused odd index, otherwise, the next term will be the next unused even index.

d. We can construct a similar series to above that diverges. Begin by adding odd index terms until the sum is greater than 2. Add the first even index term. Continue adding odd index terms until the sum is greater than 3. Add the second even index term. Repeat by adding odd index terms until the sum is greater than n, then adding the $(n-1)$-th even index term.

Exercise 0.5.5 Show that if a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then any rearrangement of the series is still convergent and converges to the same limit. *Hint:* For any $\epsilon > 0$, there exists N such that $\sum_{n=N+1}^{\infty} < \epsilon$. For any rearrangement $\sum_{k=1}^{\infty} b_n$ of the series, there exists M such that all of $a_1, \ldots, a_N$ appear among $b_1, \ldots, b_M$. Show that $\left|\sum_{n=1}^{N} a_n - \sum_{n=1}^{M} b_n\right| < \epsilon$.

SOLUTION
Let (b_n) be a rearrangement of (a_n) and $\sum_{n=1}^{\infty} = s$ converges absolutely, meaning $\sum_{n=1}^{\infty} a_n$ converges. We will first show that (b_n) converges absolutely.

For $\epsilon > 0$ we can choose $N > 0$ large enough such that

$$s - \sum_{n=1}^{N} |a_n| < \epsilon \implies \sum_{n=N+1}^{\infty} |a_n| < \epsilon$$

Since N is finite, we can find $M > 0$ such that every term from $a_1, a_2, \ldots, a_N$ appears in $b_1, b_2, \ldots, b_M$. Formally, we will write $a_k \in \{b_n : n \in \mathbb{N}, n \leq M\}$ for all $k \leq N$. Hence,

$$\begin{aligned} \sum_{n=1}^{M} b_n &= \sum_{a_k \in \{b_1, \ldots, b_M\}, k \leq N} a_k + \sum_{a_k \in \{b_1, \ldots, b_M\}, k > N} a_k \\ &= \sum_{n=1}^{N} a_n + \sum_{a_k \in \{b_1, \ldots, b_M\}, k > N} a_n \end{aligned}$$

Therefore

$$\left|\sum_{n=1}^{M} b_n - \sum_{n=1}^{N} a_n\right| = \left|\sum_{a_k \in \{b_1, \ldots, b_M\}, k > N} a_k\right| \leq \sum_{a_k \in \{b_1, \ldots, b_M\}, k > N} |a_k| \leq \sum_{n=N+1}^{\infty} |a_n| < \epsilon$$

Let

$$s_N = \sum_{n=1}^{N} a_n, \quad t_M = \sum_{n=1}^{M} b_n$$

Since $s_N \to s$, we get $|s - s_N| < \epsilon$, giving us

$$|t_M - s| \leq |t_M - s_N| + |s_N - s| < \epsilon + \epsilon = \epsilon_0$$

Thus for $M > 0$ large enough, we get

$$\lim_{m \to \infty} t_m = s$$

0.6 Infinite Sets

Exercise 0.6.1 a. Show that the set of rational numbers is countable (i.e., that you can list all rational numbers).

b. Show that the set of $\mathbb{D}$ of finite decimals is countable.

SOLUTION

a. Let

$$A = \{(p,q) \in \mathbb{Z} \times \mathbb{N} : q > 0,\ \gcd(p,q) = 1\}.$$

Every rational number has a unique representation

$$r = \frac{p}{q}, \qquad (p,q) \in A.$$

Since $\mathbb{Z}$ and $\mathbb{N}$ are countable, the Cartesian product $\mathbb{Z} \times \mathbb{N}$ is countable. Hence A, being a subset of $\mathbb{Z} \times \mathbb{N}$, is also countable.

Define

$$f : A \to \mathbb{Q}, \qquad f(p,q) = \frac{p}{q}.$$

By uniqueness of the reduced form, f is one-to-one, and by definition every rational number is of the form p/q with $(p,q) \in A$, so f is onto. Thus f is a bijection, and therefore $\mathbb{Q}$ is countable.

b. Every finite decimal can be written in the form

$$\frac{k}{10^n},$$

where $k \in \mathbb{Z}$ and $n \in \mathbb{N} \cup \{0\}$. Define

$$g : \mathbb{D} \to \mathbb{Z} \times (\mathbb{N} \cup \{0\}), \qquad g\left(\frac{k}{10^n}\right) = (k,n),$$

where the decimal is written without trailing zeros.

This map is one-to-one. Since $\mathbb{Z} \times (\mathbb{N} \cup \{0\})$ is countable, its subset $\mathbb{D}$ is also countable.

Exercise 0.6.2 a. Show that if E is finite and has n elements, then the power set $\mathcal{P}(E)$ has 2^n elements.

b. Choose a map $f : \{a,b,c\} \to \mathcal{P}(\{a,b,c\})$, and compute for that map the set $\{x | x \notin f(x)\}$. Verify that this set is not in the image of f.

SOLUTION

a. Number the elements of E as $e_1, \dots, e_n$. To form a subset $X \subseteq E$, for each element e_i we have exactly two choices:

$$e_i \in X \qquad \text{or} \qquad e_i \notin X.$$

These choices are independent, so by the multiplication principle there are

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$$

possible choices. Moreover, each sequence of choices determines a unique subset of E, and every subset arises in this way. Hence $\mathcal{P}(E)$ has exactly 2^n elements.

b. Define $f(x) = \{x\}$. Then $\{x | x \notin f(x)\} = \emptyset$, which is not in the image of f.

Exercise 0.6.3 Show that $(-1,1)$ has the same infinity of points as the reals. *Hint*: Consider the function $g(x) = \tan(\pi x/2)$.

SOLUTION
Considering the function $g(x) = \tan(\pi x/2)$ over the domain $(-1,1)$, we get the mapping $g : (-1,1) \to \mathbb{R}$. We will now show that g is a bijection.

Calculating the derivative of g, gives us $g'(x) = \frac{\pi}{2}\sec^2(\pi x/2) > 0$ for all $x \in (-1,1)$. Since g is continuous and strictly increasing on $(-1,1)$, g is one to one.

To show that g is onto, let

$$\tan\left(\frac{\pi x}{2}\right) = y \implies \frac{\pi x}{2} = \tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Then

$$-\frac{\pi}{2} < \frac{\pi x}{2} < \frac{\pi}{2} \implies -1 < x < 1 \implies x \in (-1,1)$$

Thus showing that for every $y \in \mathbb{R}$, there exists $x \in (-1,1)$ such that $g(x) = y$, implying that g is onto and therefore a bijection.

We have thus shown that $(-1,1) \asymp \mathbb{R}$.

Exercise 0.6.4 a. How many polynomials of degree d are there with coefficients with absolute value $\leq d$?

b. Give an upper bound for the position in the list of formula 0.6.5 of the polynomial $x^4 - x^3 + 5$.

c. Give an upper bound for the position of the real cube root of 2 is the list of numbers made from the list of formula 0.6.5.

SOLUTION

a. Let $p_d(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_0$ be a polynomial of degree d, with $|a_i| \leq d$ and $a_d \neq 0$.

   To find the number of polynomials, we will consider the number of cases for each coefficient.

   For a_d,

   $$a_d \in \{-d, -d+1, \ldots, d-1, d\} \setminus \{0\}$$

   which can take $(2d)$ possible values.

   For a_i, $i < d$,

   $$a_i \in \{-d, -d+1, \ldots, d-1, d\}$$

   which can take $(2d+1)$ possible values.

   Since each coefficient is independent, by the multiplication principle, there are

   $$(2d) \times \underbrace{(2d+1) \times \cdots \times (2d+1)}_{(d)\text{ times}} = (2d)(2d+1)^d$$

   possible polynomials of degree d.

b. Let P_d be the number of polynomials of degree d or lower, with coefficients with absolute value $\leq d$ and p_i be the polynomial with index position i. Then using similar reasoning to above, $P_d = (2d+1)^{d+1}$.

   While $x^4 - x^3 + 5$ is a polynomial of degree 4, the coefficient of x^0 is 5, hence it must be within the first P_5 polynomials, thus the upper bound for the position of the polynomial $x^4 - x^3 + 5$ is $P_5 = 11^6$.

c. Similarly, $x^3 - 2$ will be within the first P_3 polynomials. Since each polynomial would have at most 3 roots, we will have an upper bound of

   $$3P_3 = 3 \times 7^4$$

   for the position in the list of numbers made from the list of formula 0.6.5.

Exercise 0.6.5 Let $f : A \to B$ and $g : B \to A$ be one to one. We will sketch how to construct a mapping $h : A \to B$ that is one to one and onto.

Let an (f, g)-chain be a sequence consisting alternately of elements of A and B, with elements following an element $a \in A$ being $f(a) \in B$, and the element following an element $b \in B$ being $g(b) \in A$.

a. Show that every (f, g)-chain can be uniquely continued forever to the right, and to the left can

   1. either be uniquely continued forever, or
   2. can be continued to an element of A that is not in the image of g, or
   3. can be continued to an element of B that is not in the image of f.

b. Show that every element of A and every element of B is an element of a unique such maximal (f, g)-chain.

c. Construct $h : A \to B$ by setting

$$h(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to a maximal chain of type 1 or 2 above,} \\ g^{-1}(a) & \text{if } a \text{ belongs to a maximal chain of type 3.} \end{cases}$$

   Show that $h : A \to B$ is well defined, one to one, and onto.

d. Take $A = [-1, 1]$ and $B = (-1, 1)$. It is surprisingly difficult to write a mapping $h : A \to B$. Take $f : A \to B$ defined by $f(x) = x/2$, and $g : B \to A$ defined by $g(x) = x$.

   What elements of $[-1, 1]$ belong to chains of type 1, 2, 3?

   What map $h : [-1, 1] \to (-1, 1)$ does the construction in part c give?

SOLUTION

a. We first show that every (f, g)-chain can be uniquely continued forever to the right.

   Suppose $a \in A$. Since $f : A \to B$ is a function, $f(a) \in B$ exists uniquely. Likewise, since $g : B \to A$ is a function, $g(f(a)) \in A$ exists uniquely. Repeating this argument inductively gives the infinite chain $a,\ f(a),\ g(f(a)),\ f(g(f(a))),\ \ldots$ Thus every (f, g)-chain can be uniquely continued forever to the right.

   We now consider continuation to the left. Let

$$A^* = A \setminus g(B), \qquad B^* = B \setminus f(A).$$

   Suppose first that the current element of the chain is $a \in A$. To continue one step to the left we require an element $b \in B$ satisfying $g(b) = a$. If $a \in g(B)$, then such a b exists. Since g is one to one, this b is unique. If instead $a \in A^*$, then no such b exists, and the chain terminates on the left at a. This gives case (2).

   Similarly, suppose the current element is $b \in B$. To continue one step to the left we require an element $a \in A$ satisfying $f(a) = b$. If $b \in f(A)$, then such an a exists and is unique since f is one to one. If instead $b \in B^*$, then no such a exists, and the chain terminates on the left at b. This gives case (3).

   Finally, if neither of the above situations ever occurs, then every element encountered has a unique predecessor. Hence the chain can be uniquely continued indefinitely to the left, giving case (1). Since exactly one of these three possibilities must occur, every (f, g)-chain satisfies exactly one of the stated cases.

b. Let $x \in A \cup B$. We construct the maximal (f,g)-chain containing x.

Starting from x, repeatedly apply f and g to extend the chain to the right. By part (a), this can be done uniquely forever. Similarly, repeatedly extend the chain to the left whenever a predecessor exists. Again by part (a), each predecessor, when it exists, is unique. This process either continues forever or terminates at an element of A^* or B^*. Hence every element of $A \cup B$ belongs to a maximal (f,g)-chain.

It remains to show uniqueness. Suppose x belongs to two maximal chains C_1 and C_2. Since the predecessor and successor of every element are uniquely determined whenever they exist, both chains must have exactly the same elements to the left and to the right of x. Thus $C_1 = C_2$. Therefore every element of A and every element of B belongs to a unique maximal (f,g)-chain.

c. Firstly, we will show that the function is well defined. If a belongs to a maximal chain of type 1 or 2, each element of a would have a subsequent term since every sequence can be uniquely continued forever to the right. Thus, the term $f(a)$ exists, thus there exists an image for a.

If a belongs to a maximal chain of type 3, then the first term of the chain belongs to the set B. Since every a has a preceding term, which is equivalent to $g^{-1}(a)$. Thus the inverse exists.

Since every term has an image, h is well defined.

h is one to one. Suppose we have $a_1, a_2 \in A$, such that $h(a_1) = h(a_2)$ but $a_1 \neq a_2$. If both a_1 and a_2 belong to the a maximal chain of type 1 or 2, then $h(a_1) = f(a_1)$ and $h(a_2) = f(a_2)$, implying $f(a_1) = f(a_2)$, which a contradiction since f is one to one. Similarly, if both a_1 and a_2 belong to the a maximal chain of type 3, then $h(a_1) = g^{-1}(a_1)$ and $h(a_2) = g^{-1}(a_2)$, implying that $g^{-1}(a_1) = g^{-1}(a_2)$. This is a contradiction as g is one to one, hence g^{-1} is one to one.

Lastly, we have to consider the case where $h(a_1) = f(a_1)$ and $h(a_2) = g^{-1}(a_2)$. Let $h(a_1) = h(a_2)$, then $f(a_1) = g^{-1}(a_2)$. Since g is one to one, applying g on both sides gives us $g \circ f(a_1) = a_2$. However, this means that both a_1 and $g \circ f(a_1) = a_2$ belong to the same chain, which leads to a contradiction.

Lastly, we will show that h is onto. We need to show that for every $b \in B$, there exists $a \in A$ such that $b = h(a)$. We use the fact that if a is in the chain, then $f(a)$ and $g^{-1}(a)$ (if it exists) are both in the chain. Hence we can construct the inverse

$$h^{-1}(b) = \begin{cases} f^{-1}(b) & \text{if } b \text{ belongs to a maximal chain of type 1 or 2,} \\ g(b) & \text{if } b \text{ belongs to a maximal chain of type 3} \end{cases}$$

If $b \in B$ is from chains of type 1 and 2, then $b \in f(A)$ so $b = f(a) = h(a)$. Similarly, if b is from chains of type 3, then $h(g(b)) = g^{-1}(g(b)) = b$. Thus having a preimage and hence onto

d. Note that $f(A) = f([-1,1]) = [-1/2, 1/2]$ and $g(B) = g((-1,1)) = (-1,1)$, giving us

$$A^* = A \setminus g(B) = \{-1, 1\} \quad \text{and} \quad B^* = B \setminus f(A) = (-1, -1/2) \cup (1/2, 1)$$

For elements in A^*, there does not exist $g^{-1}(A^*)$, hence belonging to chains of type 2. With the first terms being ± 1, all terms in the chain are in the form of $\pm 2^{-n}$ for $n \in \mathbb{N}$; proving this is trivial and hence left as an exercise for the reader.

The starting elements of the chains of type 3 are B^*, and hence the elements from A that are in this chain are $g(B^*) = B^*$. Since $g \circ f(x) = \frac{x}{2}$, the subsequent terms in the chain come from the set

$$\bigcup_{n=0}^{\infty} [(-2^{-n}, -2^{-n-1}) \cup (2^{-n-1}, 2^{-n})] = [-1,1] \setminus \bigcup_{n=0}^{\infty} \{\pm 2^{-n}\}$$

Let S_n be the set of numbers from A that belong to a maximal chain of type n. Since $S_2 \cup S_3 = A$, we can conclude that there are no elements that belong to chains of type 1.

Using part c, we can construct $h : A \to B$ that is one to one and onto

$$h(x) = \begin{cases} x/2 & x \in \{\pm 2^{-n} : n \in \mathbb{N}_0\} \\ x & \text{else} \end{cases}$$

Exercise 0.6.6 Show that the points of the circle $\left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid x^2 + y^2 = 1 \right\}$ have the same infinity of elements as $\mathbb{R}$. *Hint*: This is easy if you use Bernstein's theorem (Exercise 0.6.5)

SOLUTION

We will use Bernstein's Theorem from Exercise 0.6.5, by firstly finding to injective maps.

Let us define

$$S = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid x^2 + y^2 = 1 \right\}$$

We will construct the map $f : S \to \mathbb{R}$ as follows

$$f(x, y) = \begin{cases} x + 1 & y \geq 0 \\ x - 1 & y < 0 \end{cases}$$

f is one to one. Suppose we have $(x_1, y_1), (x_2, y_2) \in S$ such that $(x_1, y_1) \neq (x_2, y_2)$ but $f(x_1, y_1) = f(x_2, y_2)$. If both $y_1, y_2 \geq 0$ or $y_1, y_2 < 0$, $x_1 \neq x_2$; otherwise both points are the same point given that $y = \sqrt{1 - x^2}$ is one to one. If $y_1 \geq 0$ and $y_2 < 0$, then $f(x_1, y_1) = f(x_2, y_2)$ implies $x_1 + 1 = x_2 - 1$ which is not possible. The two x coordinates must differ by 2, which is only possible if $x_1 = -1$ and $x_2 = 1$. However, this results in $y_1 = y_2 = 0$, leading to a contradiction on $y_2 < 0$.

Now, we construct the map $g : \mathbb{R} \to S$ as follows

$$g(x) = \big(\cos(\tan^{-1} x), \sin(\tan^{-1} x)\big) = \left(\frac{1}{\sqrt{1 + x^2}}, \frac{x}{\sqrt{1 + x^2}} \right)$$

g is clearly injective as $\tan^{-1}$ is injective and, sin and cos is also injective. By the composition of injective functions, g is also injective.

Since we have $f : S \to \mathbb{R}$ and $g : \mathbb{R} \to S$ that are both one to one, we can use Bernstein's Theorem to construct $h : S \to \mathbb{R}$ that is both one to one and onto, showing that $S \asymp \mathbb{R}$.

Exercise 0.6.7 a. Show that $[0, 1) \times [0, 1)$ has the same cardinality as $[0, 1)$.

b. Show that $\mathbb{R}^2$ has the same infinity as $\mathbb{R}$.

c. Show that $\mathbb{R}^n$ has the same infinity as $\mathbb{R}$.

SOLUTION

Consider the following notation

$$a = 0.\overline{a_1 a_2 a_3 \ldots} = \sum_{n=1}^{\infty} \frac{a_n}{10^n}$$

To remove ambiguity, all $a \in [0, 1)$ will be expressed in the decimal expansion with recurring 9s. For example, $a = \frac{1}{2}$ will be expressed as $0.4999\ldots$ and not $0.5000\ldots$.

Define

$$S = \{(x, y) : x, y \in [0, 1)\}$$

We will now begin the proof. We will define $f_0 : S \to [0, 1)$ as follows:

$$f_0(0.\overline{a_1 a_2 a_3 \ldots}, 0.\overline{b_1 b_2 b_3 \ldots}) = 0.\overline{a_1 b_1 a_2 b_2 a_3 b_3 \ldots}$$

Injective: Let $(a, b), (c, d) \in S$, such that $(a, b) \neq (c, d)$. We will show that $f_0(a, b) \neq f_0(c, d)$.

$$f_0(a, b) = 0.\overline{a_1 b_1 a_2 b_2 \ldots}, \quad f_0(c, d) = 0.\overline{c_1 d_1 c_2 d_2 \ldots}$$

Since $(a, b) \neq (c, d)$ we can find either $a_n \neq c_n$ or $b_n \neq d_n$, showing that the images are not the same. All values are in decimal expansions with recurring 9s, ensuring that there are not two elements in the domain that will map to the same image.

Define $f_1 : [0, 1) \to S$ as follows: $f_1(x) = (x, x)$, which is trivially injective. Thus we can construct $f : S \to [0, 1)$ that is bijective by Bernstein's Theorem.

It is sufficient to find a bijection $g : [0,1) \to \mathbb{R}$, as it allows us to construct $h : \mathbb{R} \to \mathbb{R}^2$, as follows:

$$h(x) = (g \times g) \circ f^{-1} \circ g^{-1}(x)$$

which is bijective as it is the composition of bijective functions.

Finding a bijection g is trivial as we can apply Bernstein's Theorem on $g_0 : [0,1) \to \mathbb{R}$ with $g_0(x) = x$ and $g_1 : \mathbb{R} \to [0,1)$ with $g_1(x) = \frac{1}{\pi}\tan^{-1} x + \frac{1}{2}$. Proving that these are one to one are trivial.

Lastly, we can use induction to map $\mathbb{R}$ to $\mathbb{R}^n$. Suppose we have a function mapping $\mathbb{R}$ to $\mathbb{R}^{n-1}$. It is sufficient to construct a function mapping $\mathbb{R}^{n-1}$ to $\mathbb{R}^n$.

Using h from the previous part, we can construct $h*$,

$$h(x) = (h_1(x), h_2(x)) \quad \text{then} \quad h^*(x_1, x_2, \dots, x_{n-1}) = (h_1(x_1), h_2(x_1), x_2, \dots, x_{n-1})$$

This is a bijection as it is the composition of bijective functions, thus $\mathbb{R} \asymp \mathbb{R}^n$

Exercise 0.6.8 Show that the power set $\mathcal{P}(\mathbb{N})$ has the same cardinality as $\mathbb{R}$.

SOLUTION

Construct the following set

$$A = \{(a_1, a_2, a_3, \dots) : a_n \in \{0,1\}\}$$

We will now show that $\mathcal{P}(\mathbb{N}) \asymp A$. We define the map $f : \mathcal{P}(\mathbb{N}) \to A$ as follows

$$f(X) = (a_1, a_2, a_3, \dots) \quad \text{where} \quad a_n = \begin{cases} 1 & n \in X \\ 0 & n \notin X \end{cases}$$

Injective: Suppose we have $f(X) = (a_1, a_2, a_3, \dots)$ and $g(Y) = (b_1, b_2, b_3, \dots)$, with $X \neq Y$. Then there exists $n \in \mathbb{N}$, such that $n \in X$ but $n \notin Y$, then $a_n = 1 \neq b_n = 0$. Thus f is one to one. **Surjective:** Given any sequence $(a_1, a_2, a_3, \dots) \in A$, we define $X = \{n \in \mathbb{N} : a_n = 1\}$, then $f(X) = (a_1, a_2, a_3, \dots)$.

Hence, we have shown that f is bijective and $\mathcal{P}(\mathbb{N}) \asymp A$.

Next,we will show that $A \asymp \mathbb{R}$ using Bernstein's Theorem in Exercise 0.6.5. We can construct a map $g_0 : A \to [0,1]$ as follows

$$g_0(a_1, a_2, a_3, \dots) = \sum_{n=1}^{\infty} \frac{2a_n}{3^n}$$

Injective: Suppose $(a_1, a_2, \dots) \neq (b_1, b_2, \dots)$. Let k be the first index where they differ, i.e. $a_k \neq b_k$ but $a_i = b_i$ for all $i < n$. Without loss of generality, $a_k = 0$ and $b_k = 1$. Then

$$g_0(b_1, b_2, \dots) - g_0(a_1, a_2, \dots) = \frac{2}{3^k} + \sum_{n>k} \frac{2(b_n - a_n)}{3^n}$$

Since $|b_n - a_n| \leq 1$, the tail satisfies

$$\left| \sum_{n>k} \frac{2(b_n - a_n)}{3^k} \right| \leq \sum_{n>k} \frac{2}{3^n} = \frac{1}{3^k}$$

Hence, we get

$$g_0(b_1, b_2, \dots) - g_0(a_1, a_2, \dots) \geq \frac{2}{3^k} - \frac{1}{3^k} > 0$$

Thus showing that g_0 is injective

We will now construct $g_1 : [0,1] \to A$ as follows

$$g_1(x) = (a_1, a_2, a_3, \dots) \quad \text{such that} \quad X = \sum_{n=1}^{\infty} \frac{a_n}{2^n}$$

For this map, $(a_n) \in A$ will be expressed in the form of recurring 0's, i.e. there does not exist $n \in \mathbb{N}$ such that $a_i = 1$ for all $i > n$. Suppose $x, y \in [0,1]$ such that $x \neq y$. Since sequences of recurring 1's are excluded, we get

$$g_1(x) = (a_1, a_2, a_3, \dots) = (b_1, b_2, b_3, \dots) = g_1(y), \quad \text{where} \quad a_n = b_n \ \forall n \in \mathbb{N}$$

By Bernstein's Theorem, we can construct $g : A \to [0,1]$ that is bijective, showing that $A \asymp [0,1]$.

Lastly, we can map $h : [0,1] \to \mathbb{R}$ that is bijective, by using Bernstein's Theorem. We have $h_0 : [0,1] \to \mathbb{R}$ with $h_0(x) = x$ and $h_1 : \mathbb{R} \to [0,1]$ with $h_1(x) = \frac{1}{\pi} \tan^{-1} x + \frac{1}{2}$. Proving that h_0 and h_1 is injective and hence h is bijective is trivial and hence left as an exercise for the reader. Thus, $[0,1] \asymp \mathbb{R}$.

Combining f, g and h, we get $\mathcal{P}(\mathbb{N}) \asymp A \asymp [0,1] \asymp \mathbb{R}$.

Exercise 0.6.9 Is it possible to list (all) the rationals in $[0,1]$, written as decimals, so that the entries on the diagonal also give a rational number?

SOLUTION

This is possible. Consider the enumeration of r_n with the condition of r_n having its n-th digit being zero. All r_n will be in the form of trailing 0's instead of trailing 9's. I.e. 0.14 will be expressed as $0.1400\dots$ instead of $0.1399\dots$ and would have a total of 2 digits.

Following this construction, $r_1 = 0.\overline{a_1 a_2 a_3 \dots}$ since it has the digit zero on the first digit. $r_2 = \overline{b_0 0 b_2 b_3 \dots}$ since it has the digit zero on the second digit. Repeating this process, the number along the diagonal would be 0.0000 which is a rational number.

0.7 Complex Numbers

Exercise 0.7.1 If z is the complex number $a + ib$, which of the following are synonyms?

modulus of z	$\lvert z\rvert$	$\bar{z}$	Re z
absolute value of z	Im z	complex conjugate of z	a
b	real part of z	imaginary part of z	

SOLUTION

$$\text{modulus of } z \equiv |z| \equiv \text{ absolute value of } z$$

$$\bar{z} \equiv \text{ complex conjugate of } z$$

$$\text{Re } z \equiv a \equiv \text{ real part of } z$$

$$\text{Im } z \equiv b \equiv \text{ imaginary part of } z$$

Exercise 0.7.2 If $z = a + ib$, which of the following are real numbers?

$\lvert z\rvert$	Re z	$\sqrt{z\bar{z}}$	$\bar{z}$
a	Im z	b	$z + \bar{z}$
modulus of z	real part of z	imaginary part of z	$z - \bar{z}$

SOLUTION

The following elements are real numbers

$$|z|, \quad \text{Re } z, \quad \sqrt{z\bar{z}}, \quad a, \quad \text{Im } z, \quad b, \quad z + \bar{z}, \quad \text{modulus of } z, \quad \text{real part of } z, \quad \text{imaginary part of } z$$

Exercise 0.7.3 Find the absolute value and argument of each of the following complex numbers:

$$\text{a. } 2+4i \quad \text{b. } (3+4i)^{-1} \quad \text{c. } (1+i)^5 \quad \text{d. } 1+4i$$

SOLUTION

a. $|2+4i| = \sqrt{2^2+4^2} = 2\sqrt{5}, \quad \arg(2+4i) = \tan^{-1}\left(\frac{4}{2}\right) = \tan^{-1} 2.$

b. $|(3+4i)^{-1}| = |3+4i|^{-1} = \sqrt{3^2+4^2}^{-1} = 1/2, \quad \arg((3+4i)^{-1}) = -\arg(3+4i) = -\tan^{-1}\left(\frac{4}{3}\right)$

c. $|(1+i)^5| = |1+i|^5 = \sqrt{2}^5 = 4\sqrt{2}, \quad \arg((1+i)^5) = 5\arg(1+i) = 5\tan^{-1} 1 = \frac{5\pi}{4}$

d. $|1+4i| = \sqrt{1^2+4^2} = \sqrt{17}, \quad \arg(1+4i) = \tan^{-1} 4$

Exercise 0.7.4 Find the modulus and polar angle of the following complex numbers:

$$\text{a. } 3+2i \quad \text{b. } (1-i)^4 \quad \text{c. } 2+i \quad \text{d. } \sqrt[7]{3+4i}$$

SOLUTION

a. $|3+2i| = \sqrt{3^2+2^2} = \sqrt{13}, \quad \arg(3+2i) = \tan^{-1}\left(\frac{2}{3}\right).$

b. $|(1-i)^4| = |1-i|^4 = 4, \quad \arg((1-i)^4) = 4\arg(1-i) = 4\tan^{-1}(-1) = -\pi \equiv \pi \pmod{2\pi}.$

c. $|2+i| = \sqrt{2^2+1^2} = \sqrt{5}, \quad \arg(2+i) = \tan^{-1}\left(\frac{1}{2}\right).$

d. $|\sqrt[7]{3+4i}| = |3+4i|^{1/7} = \sqrt{3^2+4^2}^{1/7} = \sqrt[7]{5}, \quad \arg(\sqrt[7]{(3+4i)}) = \frac{1}{7}\tan^{-1}\left(\frac{4}{3}\right).$

Exercise 0.7.5 Verify the nine rules for addition and multiplication of complex numbers. Statements 5 and 9 are the only ones that are not immediate.

SOLUTION

To recall, the nine rules, for $z_1, z_2, z_3 \in \mathbb{C}$ are:

$$\begin{cases} 1.\ (z_1+z_2)+z_3 = z_1+(z_2+z_3) & \text{Addition is associative.} \\ 2.\ z_1+z_2 = z_2+z_1 & \text{Addition is commutative.} \\ 3.\ z+0 = z & \text{0 is an additive identity.} \\ 4.\ (a+ib)+(-a-ib) = 0 & \text{Existence of an additive inverse.} \\ 5.\ (z_1z_2)z_3 = z_1(z_2z_3) & \text{Multiplication is associative.} \\ 6.\ z_1z_2 = z_2z_1 & \text{Multiplication is commutative.} \\ 7.\ 1z = z & \text{1 is a multiplicative identity.} \\ 8.\ (a+ib)\left(\frac{a}{a^2+b^2} - i\frac{b}{a^2+b^2}\right) = 1 & \text{Existence of a multiplicative inverse.} \\ 9.\ z_1(z_2+z_3) = z_1z_2+z_1z_3 & \text{Multiplication is distributive over addition.} \end{cases}$$

Rules 1 to 4 are trivial since the addition of complex numbers is equivalent to the addition under $\mathbb{R}^2$.

We will show the multiplicative properties. Rule 6 is true since $\mathbb{R}$ also satisfies the multiplicative commutative property

$$(a_1+ib_1)(a_2+ib_2) = a_1a_2 - b_1b_2 + i(a_1b_2+a_2b_1) = a_2a_1 - b_2b_1 + i(b_2a_1+b_1a_2) = (a_2+ib_2)(a_1+ib_1).$$

Rule 7 is trivially true as

$$1(a+ib) = 1a + i(1b) = a+ib.$$

Rule 8 is also true by simply expanding the expression. This is trivial and hence left as an exercise for the reader.

We have left Rules 5 and 9 to the last since they are more complicated. For Rule 5, we expand the expressions as follows:

$$\begin{aligned}(z_1z_2)z_3 &= \Big[(a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)\Big](a_3 + ib_3)\\ &= \Big[(a_1a_2 - b_1b_2)a_3 - (a_1b_2 + a_2b_1)b_3\Big] + i\Big[(a_1a_2 - b_1b_2)b_3 + (a_1b_2 + a_2b_1)a_3\Big]\\ &= (a_1a_2a_3 - a_3b_1b_2 - a_1b_2b_3 - a_2b_1b_3) + i(a_1a_2b_3 - b_1b_2b_3 + a_1a_3b_2 + a_2a_3b_1),\\ z_1(z_2z_3) &= (a_1 + ib_1)\Big[(a_2a_3 - b_2b_3) + i(a_2b_3 + a_3b_2)\Big]\\ &= \Big[a_1(a_2a_3 - b_2b_3) - b_1(a_2b_3 + a_3b_2)\Big] + i\Big[a_1(a_2b_3 + a_3b_2) + b_1(a_2a_3 - b_2b_3)\Big]\\ &= (a_1a_2a_3 - a_1b_2b_3 - a_2b_1b_3 - a_3b_1b_2) + i(a_1a_2b_3 + a_1a_3b_2 + a_2a_3b_1 - b_1b_2b_3).\end{aligned}$$

which is equivalent since addition over $\mathbb{R}$ is also associative.

Lastly, Rule 9 follows a similar approach:

$$\begin{aligned}z_1(z_2 + z_3) &= (a_1 + ib_1)\big[(a_2 + a_3) + i(b_2 + b_3)\big]\\ &= \Big[a_1(a_2 + a_3) - b_1(b_2 + b_3)\Big] + i\Big[a_1(b_2 + b_3) + b_1(a_2 + a_3)\Big]\\ &= (a_1a_2 + a_1a_3 - b_1b_2 - b_1b_3) + i(a_1b_2 + a_1b_3 + a_2b_1 + a_3b_1),\\ z_1z_2 + z_1z_3 &= \Big[(a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)\Big] + \Big[(a_1a_3 - b_1b_3) + i(a_1b_3 + a_3b_1)\Big]\\ &= (a_1a_2 + a_1a_3 - b_1b_2 - b_1b_3) + i(a_1b_2 + a_1b_3 + a_2b_1 + a_3b_1).\end{aligned}$$

Exercise 0.7.6 a. Solve $x^2 + ix = -1$. b. Solve $x^4 + x^2 + 1 = 0$

SOLUTION

Moving terms to one side gives us $x^2 + ix + 1 = 0$. By quadratic formula,

$$x = \frac{-i \pm \sqrt{i^2 - 4(1)(1)}}{2} = \frac{-i \pm \sqrt{5}i}{2}.$$

Using quadratic formula,

$$\begin{aligned}x^2 &= \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2}\\ &= \frac{-1 \pm \sqrt{3}\,i}{2}\\ &= \cos\frac{2\pi}{3} \pm i\sin\frac{2\pi}{3}\\ &= e^{\pm 2\pi i/3},\end{aligned}$$

so

$$x = e^{i\theta}, \qquad \theta = \pm\frac{\pi}{3},\ \pm\frac{2\pi}{3}.$$

Exercise 0.7.7 Describe the set of all complex number $z = x + iy$ such that

a. $|z-u|+|z-v| = c$ for $u, v \in \mathbb{C}$ and $c \in \mathbb{R}$ b. $|z| < 1 - \text{Re } z$

SOLUTION

a. By properties of conic sections, this set represents an ellipse with the foci of u and v on the Argand Diagram. However, we will need to prove this rigorously with the correct conditions. Let $S = \{z \in C : |z-u|+|z-v| = c \text{ for } u, v \in \mathbb{C} \text{ and } c \in R\}$.

By Triangle Inequality,

$$|z-u|+|z-v| \geq |u-v|$$

Hence we have to consider the 3 cases, $c < |u-v|$, $c = |u-v|$ and $c > |u-v|$.

Case 1: Suppose $c < |u-v|$. By triangle Inequality, we get $|z-u|+|z-v| > c$, which suggest there does not exists any $z \in \mathbb{C}$ that satisfy the set. Hence, $S = \emptyset$.

Case 2: Suppose $c = |u-v|$. The equality is achieved in the Triangle Inequality, hence, z, u and v must be collinear with the condition that $z-u$ and $v-z$ must be positive scalar multiples of each other. Hence, S is a line on the Argand Diagram connecting u and v. To express this formally, we get

$$S = \{z = u + t(v-u) : t \in [0,1]\}$$

Case 3: Suppose $c > |u-v|$. This case is the most difficult to prove. Before we begin, let us recall the geometric representation of addition and multiplication of complex numbers. Substituting z with $z+c$ caused the point in the Argand diagram to move Im c units downwards and Re c units leftwards. Substituting z with $ze^{i\theta}$ causes the point to rotate θ radians clockwise.

Let σ_1 be a transformation substituting z with $z+w$, $(z \to z+w)$, where $w = (u+v)/2$, this results in a new set

$$\begin{aligned} S_1 = \sigma_1 S &= \{z \in \mathbb{C} : \left|z - \frac{v}{2} + \frac{u}{2}\right| + \left|z - \frac{u}{2} + \frac{v}{2}\right| = c \text{ for } u, v \in \mathbb{C}, c \in \mathbb{R}\} \\ &= \{z \in \mathbb{C} : |z-u_1| + |z-v_1| = c \text{ for } u, v \in \mathbb{C}, c \in \mathbb{R}\} \end{aligned}$$

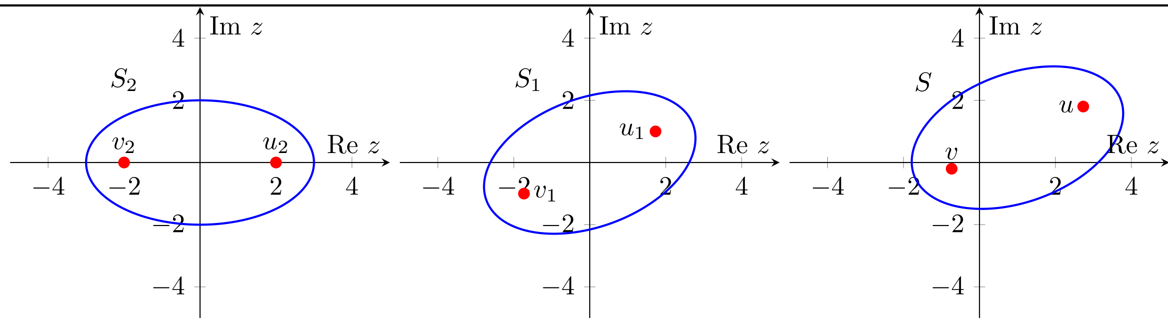
Without loss of generality, suppose Re $u_1 >$ Re v_1. Let $\theta = \arg(u_1 - v_1)$ and σ_2 be the transformation substituting z with $ze^{i\theta}$, ie,$(z \to ze^{i\theta})$. This gives us a new set

$$\begin{aligned} S_2 = \sigma_2 S_1 = &\{z \in \mathbb{C} : \left|ze^{i\theta} - u_1\right| + \left|ze^{i\theta} - v_1\right| = c \text{ for } u, v \in \mathbb{C}, c \in \mathbb{R}\} \\ &\{z \in \mathbb{C} : |e^{i\theta}|\left|z - u_1 e^{-i\theta}\right| + |e^{i\theta}|\left|z - v_1 e^{-i\theta}\right| = c \text{ for } u, v \in \mathbb{C}, c \in \mathbb{R}\} \\ &\{z \in \mathbb{C} : |z-k| + |z+k| = c \text{ for } u, v \in \mathbb{C}, c \in \mathbb{R}\} \end{aligned}$$

Since both points now lie on real axis and are equidistant to the origin, we can substitute them for k and $-k$.

Now we have to show that this set is an ellipse.

$$\begin{aligned} |z-k| + |z+k| &= c \\ \sqrt{(x+k)^2 + y^2} + \sqrt{(x-k)^2 + y^2} &= c \\ \sqrt{(x+k)^2 + y^2} &= c - \sqrt{(x-k)^2 + y^2} \\ (x+k)^2 + y^2 &= c^2 - 2c\sqrt{(x-k)^2 + y^2} + (x-k)^2 + y^2 \\ 4kx &= c^2 - 2c\sqrt{(x-k)^2 + y^2} \\ 2c\sqrt{(x-k)^2 + y^2} &= c^2 - 4kx \\ 4c^2\left((x-k)^2 + y^2\right) &= (c^2 - 4kx)^2 \\ 4c^2(x^2 - 2kx + k^2 + y^2) &= c^4 - 8c^2kx + 16k^2x^2 \\ 4c^2x^2 + 4c^2y^2 + 4c^2k^2 &= c^4 + 16k^2x^2 \\ 4(c^2 - 4k^2)x^2 + 4c^2y^2 &= c^2(c^2 - 4k^2). \end{aligned}$$



Lastly, with reference to the diagram, we find the inverse of the transformation, finding a rotated ellipse with the foci of u and v.

b. Let $z = a + ib$, then

$$\begin{aligned} |z| &< 1 - \operatorname{Re} z \\ \sqrt{a^2+b^2} &< 1 - a \implies a < 1, \\ a^2 + b^2 &< (1-a)^2 \\ b^2 &< 1 - 2a \\ a &< \frac{1-b^2}{2}. \end{aligned}$$

This results in the region to the left of the parabola

$$a = \frac{1-b^2}{2},$$

which has focus $(0, 0)$ and directrix $a = 1$, as shown below.

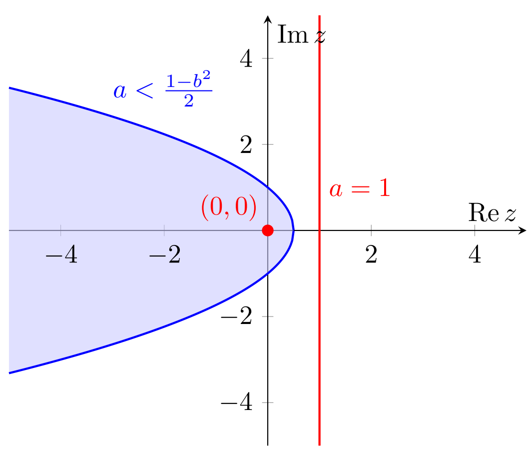


Figure 1: Region satisfying $\{z \in \mathbb{C} : |z| < 1 - \operatorname{Re} z\}$.

Exercise 0.7.8 a. Solve $x^2 + 2ix - 1 = 0$. b. Solve $x^3 - x^2 - x = 2$.

SOLUTION

a. Using the quadratic formula, we get

$$x = \frac{-2i \pm \sqrt{-4 - 4(-1)(1)}}{2} = -i$$

b. Moving all terms to one side gives us $x^3 - x^2 - x - 2 = 0$. Since every odd polynomial has at least one real root, we can use Intermediate Value Theorem to locate the first root.

$$f(1) = 1 - 1 + 1 - 2 = -3 \quad \text{and} \quad f(3) = 27 - 9 - 3 - 2 = 13$$

Hence we conclude there is a root on the interval $(1, 3)$. By repeating this process, we can eventually show that $(x - 2)$ is a factor.

$$(x^3 - x^2 - x - 2) = x^3 - 2x^2 + x^2 - 2x + x - 2 = (x - 2)(x^2 + x + 1)$$

By quadratic formula, we get

$$x = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$$

Hence the roots are

$$x = 2, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

Exercise 0.7.9 Solve the following equation where x, y are complex numbers:

$$\text{a. } x^2 + ix + 2 = 0 \qquad \text{b. } x^4 + x^2 + 2 = 0 \qquad \text{c. } \begin{cases} ix - (2+i)y = 3 \\ (1+i)x + iy = 4 \end{cases}$$

SOLUTION

a. Using quadratic formula,

$$x = \frac{-i \pm \sqrt{i^2 - 4(1)(2)}}{2} = \frac{-i \pm \sqrt{-9}}{2} = -\frac{i}{2} \pm \frac{3i}{2}$$

Hence, $x = i \quad \text{or} \quad x = -2i$.

b. Using quadratic formula, we can solve for x^2,

$$x^2 = \frac{-1 \pm \sqrt{1 - 4(1)(2)}}{2} = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}i$$

Hence by solving, we get

$$x = \sqrt{\frac{1}{2} \pm \frac{\sqrt{7}}{2}i} \quad \text{OR} \quad -\sqrt{\frac{1}{2} \pm \frac{\sqrt{7}}{2}i}$$

c.

$$\begin{cases} ix - (2+i)y = 3(1) \\ (1+i)x + iy = 4(2) \end{cases}$$

From equation (1),

$$\begin{aligned} ix(-i) - (2+i)y(-i) &= -3i \\ x + (2i - 1)y &= -3i \\ (1+i)x + (2i-1)(1+i)y &= -3i(1+i) \\ (1+i)x + (-3+i)y &= 3 - 3i. \end{aligned} \tag{3}$$

Subtracting (3) from (2),

$$3y = 4 - (3 - 3i) = 1 + 3i$$

so

$$y = \frac{1 + 3i}{3} = \frac{1}{3} + i.$$

Substituting into

$$x + (2i - 1)y = -3i,$$

we obtain

$$\begin{aligned} x + (2i-1)\left(\frac{1}{3} + i\right) &= -3i \\ x + \left(-\frac{7}{3} - \frac{1}{3}i\right) &= -3i \\ x &= \frac{7}{3} - \frac{8}{3}i. \end{aligned}$$

Hence, $x = \frac{7}{3} - \frac{8}{3}i, \qquad y = \frac{1}{3} + i.$

Exercise 0.7.10 Describe the set of all complex numbers $z = x + iy$ such that

a. $\bar{z} = -z$ b. $|z - a| = |z - b|$ for given $a, b \in \mathbb{C}$ c. $\bar{z} = z^{-1}$

SOLUTION

a. Substiting $z = x + iy$,

$$\begin{aligned} \overline{x + iy} &= -(x + iy) \\ x - iy &= -x - iy \end{aligned}$$

Comparing the coefficients, we get

$$\begin{cases} x = -x \\ y = y \end{cases} \implies \begin{cases} x = 0 \\ y \in \mathbb{R} \end{cases}$$

Hence the set is a vertical line along the Imaginary axis.

b. This set describes the perpendicular bisector of a and b on the Argand Diagram. We will have to show that this set is a line to prove this, since the points by definition match that of a perpendicular bisector.

Without loss of generality, we can suppose $a = k$ and $b = -k$ where $k \in \mathbb{R}$; since we can find a series of transformation similar to 0.7.7a.

$$\begin{aligned} |x + iy - k| &= |x + iy + k| \\ \sqrt{(x-k)^2 + y^2} &= \sqrt{(x+k)^2 + y^2} \\ (x-k)^2 + y^2 &= (x+k)^2 + y^2 \\ (x+k)^2 - (x-k)^2 &= 0 \\ (x + k + x - k)(x + k - x + k) &= 0 \\ 4kx &= 0 \implies x = 0, \quad y \in \mathbb{R} \end{aligned}$$

We have thus shown the set describes a line, which means that it is the perpendicular bisector of a and b.

c. Substituting $z = a + ib$, we get

$$\overline{a + ib} = (a + ib)^-1$$

To avoid the algebra involved, we will be comparing the argument and modulus of both side.

$$|\bar{z}| = |z^{-1}| \implies |z| = \frac{1}{|z|} \implies |z| = 1 \tag{1}$$

Similarly, let arg $z = \theta$. Then,

$$\arg\,(\bar{z}) = \arg\ z^{-1} \implies -\theta = -\theta \implies \theta \in [0, 2pi)$$

Hence we can conclude that the set represents a unit circle with 0 as its centre.

Note: Before beginning Exercise 0.7.11 and 0.7.12, let $z = x + iy$ and $z^2 = u + iv$, then $u = x^2 - y^2$ and $v = 2xy$.

Exercise 0.7.11 a. Describe the loci in $\mathbb{C}$ given by the following equations:

i. Re $z = 1$,

ii. $|z| = 3$.

b. What is the image of each locus under the mapping $z \mapsto z^2$?

c. What is the inverse image of each locus under the mapping $z \mapsto z^2$?

SOLUTION

a. i. This is set is a veritcal line intersecting the real axis at 1, following the equation of $x = 1$.

ii. This set is a circle of radius 3 and a center of 0, following the equation $x^2 + y^2 = 9$.

b. i. Substituting $x = 1$, we get

$$u = 1 - y^2, \quad v = 2y$$

Combining both equations to eliminate y, we get

$$v^2 = 4(1 - u)$$

which is a parabola.

ii. Since $|z^2| = |z|^2$ and $|z| = 3$, then we get $|z^2| = 9$, showing that the image is a circle with center 0 and radius 9.

c. i. Consdering the inverse of $u = 1$, then $x^2 - y^2 = 1$, which is the equation of a hyperbola.

ii. Similar to above, the image is a circle centered at the origin with radius of $\sqrt{3}$.

Exercise 0.7.12 a. Describe the loci in $\mathbb{C}$ given by the following equations:

i. Im $z = -|z|$,

ii. $\frac{1}{2}|z - i| = z$.

b. What is the image of each locus under the mapping $z \mapsto z^2$?

c. What is the inverse image of each locus under the mapping $z \mapsto z^2$?

SOLUTION

a. i. Substituting $z = x + iy$, we get

$$y = -\sqrt{x^2 + y^2} \implies (-y)^2 = x^2 + y^2 \ (y \leq 0) \implies x = 0$$

Hence the set is a non-positive imaginary axis.

ii. Comparing real and imaginary coefficients, we get

$$\begin{cases} \text{Re } \left(\frac{1}{2}|z - i|\right) = \text{Re } z \\ \text{Im } \left(\frac{1}{2}|z - i|\right) = \text{Im } z \end{cases} \implies \begin{cases} \sqrt{x^2 + (y-1)^2} = 2x \\ y = 0 \end{cases}$$

$$\implies x^2 + 1 = 4x^2 \implies x = \frac{1}{\sqrt{3}} \because x \geq 0$$

Hence the set is a single point at $\left(\frac{1}{\sqrt{3}}\right)$.

b. i. Since $z = -yi$ for $y \in \mathbb{R}_0^+$, we get $z^2 = -y^2$, which is the non positive real axis.

ii. $z^2 = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$.

c. i. Finding the inverse mapping is simple considering the argument, and hence the proof left as an exercise to the reader. The inverse map is $\{x - ix : x \in \mathbb{R}\}$.

ii. This is the points $\pm\frac{1}{\sqrt[4]{3}}$.

Exercise 0.7.13 a. Find all the cubic roots of 1.

b. Find all the 4th roots of 1.

c. Find all the 6th roots of 1.

SOLUTION
The answers to these are simple using the De Moivre's formula.

a. $1^{1/3} = (e^{2\pi k i})^{1/3}$ for $k = 0, 1, 2$. Hence, the roots are

$$1, e^{i\frac{2\pi}{3}}, e^{-i\frac{2\pi}{3}}$$

b. Similarly, the roots are ± 1 and $\pm i$.

c. Similarly, the roots are ± 1, $e^{\pm i\frac{\pi}{3}}$ and $e^{\pm i\frac{2\pi}{3}}$.

Exercise 0.7.14 Show that for any complex numbers z_1, z_2, we have $\mathrm{Im}(\bar{z_1} z_2) \leq |z_1||z_2|$.

SOLUTION
It is obvious that $\mathrm{Im}(\bar{z}_1 z_2) \geq |z_1||z_2|$ can be rewritten as $\mathrm{Im}(z_1 z_2) \geq |z_1 z_2|$ since $|z_1| = |\bar{z}_1|$.
Then we get

$$|z_1 z_2| = |\mathrm{Re}(z_1 z_2) + i\mathrm{Im}(z_1 z_2)| \geq |\mathrm{Re}(z_1 z_2)| + |\mathrm{Im}(z_1 z_2)| \geq |\mathrm{Im}(z_1 z_2)| \geq \mathrm{Im}(z_1 z_2)$$

and we are done.

Exercise 0.7.15 a. Find all the fifth roots of 1, using a formula that only includes square roots (no trigonometric function).

b. Use your formula, ruler, and compass to construct a regular pentagon.

SOLUTION

a. Note that the fifth roots of 1 satisfy the equation

$$z^5 - 1 = 0,$$

and we know that $z - 1$ is a factor since 1 is a solution.

$$z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1) = 0.$$

Excluding the solution $z = 1$, divide by z^2 to obtain

$$\begin{aligned} z^2 + z + 1 + \frac{1}{z} + \frac{1}{z^2} &= 0 \\ \left(z^2 + 2 + \frac{1}{z^2}\right) + \left(z + \frac{1}{z}\right) - 1 &= 0 \\ \left(z + \frac{1}{z}\right)^2 + \left(z + \frac{1}{z}\right) - 1 &= 0 \\ z + \frac{1}{z} &= \frac{-1 \pm \sqrt{5}}{2} \end{aligned}$$

Hence, We now solve for z.

If $z + \frac{1}{z} = \frac{-1+\sqrt{5}}{2}$, then

$$\begin{aligned} z^2 - \frac{-1+\sqrt{5}}{2} z + 1 = 0 \\ z &= \frac{\frac{-1+\sqrt{5}}{2} \pm \sqrt{\left(\frac{-1+\sqrt{5}}{2}\right)^2 - 4}}{2} \\ &= \frac{-1+\sqrt{5} \pm i\sqrt{10+2\sqrt{5}}}{4}. \end{aligned}$$

If $z + \frac{1}{z} = \frac{-1-\sqrt{5}}{2}$, then

$$\begin{aligned} z^2 - \frac{-1-\sqrt{5}}{2} z + 1 = 0 \\ z &= \frac{\frac{-1-\sqrt{5}}{2} \pm \sqrt{\left(\frac{-1-\sqrt{5}}{2}\right)^2 - 4}}{2} \\ &= \frac{-1-\sqrt{5} \pm i\sqrt{10-2\sqrt{5}}}{4}. \end{aligned}$$

Together with $z = 1$, the five fifth roots of unity are

$$1, \quad \frac{-1+\sqrt{5} \pm i\sqrt{10+2\sqrt{5}}}{4}, \quad \frac{-1-\sqrt{5} \pm i\sqrt{10-2\sqrt{5}}}{4}.$$

b. **TODO**